Diag Solutions

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1. **1.** Find two consecutive positive integers, sum of whose squares is 365.

   Answer:

$$13 and 14$$

   Solution:

   1. Let the two consecutive integers be x and $x + 1$.
   2. Set up the equation: $x^2 + (x + 1)^2 = 365$.
   3. Expand and simplify to standard form: $2x^2 + 2x - 364 = 0$, or $x^2 + x - 182 = 0$.
   4. Solve using factorization: $(x + 14)(x - 13) = 0$. Since integers are positive, $x = 13$.

   Derivation:

$$\begin{aligned} x^2 + x^2 + 2x + 1 &= 365 \\ 2x^2 + 2x - 364 &= 0 \\ x^2 + x - 182 &= 0 \\ (x + 14)(x - 13) &= 0 \\ x = 13, x + 1 &= 14 \end{aligned}$$

Quadratic Equations — Word Problems

2. **2.** Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

   Answer:

$$k = \pm 2\sqrt{6}$$

   Solution:

   1. For equal real roots, the discriminant $D = b^2 - 4ac$ must be equal to 0.
   2. Identify coefficients: $a = 2, b = k, c = 3$.
   3. Substitute into discriminant formula: $k^2 - 4(2)(3) = 0$.
   4. Solve for k: $k^2 - 24 = 0$, so $k^2 = 24$, $k = \pm\sqrt{24} = \pm 2\sqrt{6}$.

   Derivation:

$$\begin{aligned} b^2 - 4ac &= 0 \\ k^2 - 4(2)(3) &= 0 \\ k^2 &= 24 \\ k &= \pm 2\sqrt{6} \end{aligned}$$

3. **3.** The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

   Answer:

$$\text{Base} = 12cm, \text{Altitude} = 5cm$$

   Solution:

   1. Let base be x cm, then altitude is $(x-7)$ cm.
   2. Apply Pythagoras theorem: $x^2 + (x-7)^2 = 13^2$.
   3. Expand and simplify: $x^2 + x^2 - 14x + 49 = 169 \Rightarrow 2x^2 - 14x - 120 = 0$.
   4. Divide by 2: $x^2 - 7x - 60 = 0$. Factorize: $(x-12)(x+5) = 0$. Base must be positive, so $x = 12$.

   Derivation:

$$\begin{aligned} x^2 + (x-7)^2 &= 13^2 \\ 2x^2 - 14x + 49 - 169 &= 0 \\ 2x^2 - 14x - 120 &= 0 \\ x^2 - 7x - 60 &= 0 \\ (x-12)(x+5) &= 0 \end{aligned}$$

Quadratic Equations — Word Problems

4. **4.** Solve the quadratic equation $x^2 - 5x + 6 = 0$ using the quadratic formula.

   Answer:

$$x = 2, 3$$

   Solution:

   1. Identify coefficients: $a = 1, b = -5, c = 6$.
   2. Calculate the discriminant: $D = b^2 - 4ac = (-5)^2 - 4(1)(6) = 25 - 24 = 1$.
   3. Use the quadratic formula $x = \frac{-b \pm \sqrt{D}}{2a}$.
   4. Substitute values: $x = \frac{5 \pm 1}{2}$. This gives $x = \frac{6}{2} = 3$ and $x = \frac{4}{2} = 2$.

   Derivation:

$$\begin{aligned} x &= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6)}}{2(1)} \\ x &= \frac{5 \pm \sqrt{25 - 24}}{2} \\ x &= \frac{5 \pm 1}{2} \\ x &= 3, 2 \end{aligned}$$

5. **5.** A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Answer:

$$6km/h$$

Solution:

1. Let the speed of the stream be x km/h.
2. Speed upstream = $(18 - x)$ km/h; Speed downstream = $(18 + x)$ km/h.
3. Time taken upstream $T_1 = \frac{24}{18-x}$ and time taken downstream $T_2 = \frac{24}{18+x}$.
4. Set up the equation: $T_1 - T_2 = 1$.
5. Solve the resulting quadratic equation $x^2 + 48x - 324 = 0$.
6. Factorize to find $x = 6$ or $x = -54$. Since speed cannot be negative, $x = 6$.

Derivation:

$$\begin{aligned} \frac{24}{18-x} - \frac{24}{18+x} &= 1 \\ \implies 24(18 + x - 18 + x) &= (18 - x)(18 + x) \\ \implies 48x &= 324 - x^2 \\ \implies x^2 + 48x - 324 &= 0 \\ \implies (x + 54)(x - 6) &= 0 \end{aligned}$$

Quadratic Equations — Word problems involving speed, distance, and time

6. **6.** The sum of the areas of two squares is 468 m^2. If the difference of their perimeters is 24 m, find the sides of the two squares.

Answer:

$$12mand18m$$

Solution:

1. Let sides of squares be x and y $(x > y)$.
2. Perimeters are $4x$ and $4y$. Given $4x - 4y = 24$, so $x - y = 6 \implies x = y + 6$.
3. Sum of areas: $x^2 + y^2 = 468$.
4. Substitute x: $(y + 6)^2 + y^2 = 468$.
5. Expand and simplify: $y^2 + 12y + 36 + y^2 = 468 \implies 2y^2 + 12y - 432 = 0$.
6. Simplify to $y^2 + 6y - 216 = 0$ and solve for y using the quadratic formula or factoring.

Derivation:

$$y^2 + 6y - 216 = 0$$
$$\implies (y+18)(y-12) = 0$$
$$\implies y = 12 \text{ (since side ¿ 0)}, x = 12 + 6 = 18$$

Quadratic Equations — Geometric problems

7. **7.** A peacock is sitting on the top of a pillar, which is 9 m high. From a point 27 m away from the bottom of the pillar, a snake is coming to its hole at the base of the pillar. Seeing the snake, the peacock pounces on it. If their speeds are equal, at what distance from the hole is the snake caught?

   Answer:

   $$12m$$

   Solution:

   1. Let the snake be caught at distance x from the hole. The snake travels $27 - x$ distance.
   2. The peacock flies from the top of the pillar to the point of catch. By Pythagoras theorem, distance covered by peacock $= \sqrt{9^2 + x^2}$.
   3. Since speeds are equal and they start at the same time, distances are equal: $27 - x = \sqrt{81 + x^2}$.
   4. Square both sides: $(27 - x)^2 = 81 + x^2$.
   5. Expand: $729 - 54x + x^2 = 81 + x^2$.
   6. Solve for x: $54x = 648 \implies x = 12$.

   Derivation:

   $$729 - 54x = 81$$
   $$\implies 54x = 648$$
   $$\implies x = \frac{648}{54} = 12$$

   Quadratic Equations — Application of quadratic equations in geometry

8. **8.** A shopkeeper buys a number of books for 80 rupees. If he had bought 4 more books for the same amount, each book would have cost 1 rupee less. How many books did he buy?

   Answer:

   $$16\text{books}$$

   Solution:

   1. Let the number of books be x.
   2. Original cost per book $= \frac{80}{x}$.
   3. New number of books $= x + 4$. New cost per book $= \frac{80}{x+4}$.
   4. Given the difference in price is 1: $\frac{80}{x} - \frac{80}{x+4} = 1$.

5. Simplify: $80(x+4) - 80x = x(x+4)$.
6. Form quadratic: $x^2 + 4x - 320 = 0$.
7. Solve: $(x+20)(x-16) = 0$. Since $x > 0$, $x = 16$.

Derivation:

$$\begin{aligned} 80x + 320 - 80x &= x^2 + 4x \\ \implies x^2 + 4x - 320 &= 0 \\ \implies x = \frac{-4 \pm \sqrt{16 - 4(1)(-320)}}{2} &= \frac{-4 \pm 36}{2} \end{aligned}$$

Quadratic Equations — Word problems based on commercial mathematics

9. **9.** Assertion (A): The common difference of the AP $5, 2, -1, -4, ...$ is -3. Reason (R): The common difference d of an AP $a_1, a_2, a_3, ...$ is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate common difference: $d = a_2 - a_1 = 2 - 5 = -3$.
2. Verify the formula for common difference.

Derivation:

$$d = 2 - 5 = -3; \text{ Yes, } d = a_n - a_{n-1} \text{ is the definition.}$$

Arithmetic Progressions — Common Difference

10. **10.** Assertion (A): The 10th term of the AP $2, 7, 12, ...$ is 47. Reason (R): The n^{th} term of an AP is given by $a_n = a + (n-1)d$.

Answer: (A)

Solution:

1. Identify $a = 2, d = 5, n = 10$.
2. Apply $a_{10} = 2 + (10-1)5 = 2 + 45 = 47$.

Derivation:

$$a_{10} = 2 + (9)(5) = 2 + 45 = 47$$

Arithmetic Progressions — nth term of an AP

11. **11.** Assertion (A): The sum of the first 10 natural numbers is 55. Reason (R): The sum of first n natural numbers is given by $\frac{n(n+1)}{2}$.

Answer: (A)

Solution:

1. Use formula $S_n = \frac{n(n+1)}{2}$ for $n = 10$.
2. Calculate $S_{10} = \frac{10 \times 11}{2} = 55$.

Derivation:

$$S_{10} = \frac{10(11)}{2} = 5imes11 = 55$$

Arithmetic Progressions — Sum of first n terms

12. **12.** Assertion (A): The sequence $3, 3, 3, 3, ...$ is an AP. Reason (R): A sequence is an AP if the difference between consecutive terms is constant.

Answer: (A)

Solution:

1. Calculate differences: $3 - 3 = 0, 3 - 3 = 0$.
2. Since difference is constant ($d = 0$), it is an AP.

Derivation:

$$a_2 - a_1 = 0; a_3 - a_2 = 0; d = 0 \text{ (constant)}$$

Arithmetic Progressions — Definition of AP

13. **13.** Assertion (A): If $2x, x + 10, 3x + 2$ are in AP, then $x = 6$. Reason (R): If a, b, c are in AP, then $2b = a + c$.

Answer: (A)

Solution:

1. Use property $2(x + 10) = 2x + (3x + 2)$.
2. Solve for x: $2x + 20 = 5x + 2 \implies 18 = 3x \implies x = 6$.

Derivation:

$$\begin{aligned} 2(x + 10) &= 2x + 3x + 2 \\ \implies 2x + 20 &= 5x + 2 \\ \implies 18 &= 3x \\ \implies x &= 6 \end{aligned}$$

Arithmetic Progressions — Arithmetic Mean property

14. **14.** Find the 10^{th} term of the Arithmetic Progression $5, 8, 11, 14, \ldots$

Answer:

$$32$$

Solution:

1. Identify the first term $a = 5$ and the common difference $d = 8 - 5 = 3$.
2. Use the formula $a_n = a + (n - 1)d$ to calculate the 10^{th} term.

Derivation:

$$\begin{aligned} a_n &= a + (n-1)d \\ a_{10} &= 5 + (10-1)3 \\ a_{10} &= 5 + 9 \times 3 \\ a_{10} &= 5 + 27 = 32 \end{aligned}$$

Arithmetic Progressions — nth term of an AP

15. **15.** Which term of the Arithmetic Progression $21, 18, 15, \ldots$ is -81?

Answer:

$$35^{th}\text{term}$$

Solution:

1. Identify $a = 21$ and $d = 18 - 21 = -3$.
2. Set $a_n = -81$ and solve the equation $-81 = 21 + (n-1)(-3)$ for n.

Derivation:

$$\begin{aligned} -81 &= 21 - 3n + 3 \\ -81 &= 24 - 3n \\ 3n &= 24 + 81 \\ 3n &= 105 \\ n &= 35 \end{aligned}$$

Arithmetic Progressions — nth term of an AP

16. **16.** Find the sum of the first 20 terms of the Arithmetic Progression $1, 6, 11, 16, \ldots$

Answer:

$$970$$

Solution:

1. Identify $a = 1$, $d = 5$, and $n = 20$.
2. Use the sum formula $S_n = \frac{n}{2}[2a + (n-1)d]$.

Derivation:

$$\begin{aligned} S_{20} &= \frac{20}{2}[2(1) + (20-1)5] \\ S_{20} &= 10[2 + 19 \times 5] \\ S_{20} &= 10[2 + 95] \\ S_{20} &= 10 \times 97 = 970 \end{aligned}$$

Arithmetic Progressions — Sum of first n terms

17. **17.** In an Arithmetic Progression, if the first term is 7 and the common difference is 4, find the sum of the first 10 terms.

Answer:

$$250$$

Solution:

1. State the given values: $a = 7$, $d = 4$, $n = 10$.
2. Substitute these values into $S_n = \frac{n}{2}[2a + (n-1)d]$.

Derivation:

$$\begin{aligned} S_{10} &= \frac{10}{2}[2(7) + (10-1)4] \\ S_{10} &= 5[14 + 9 \times 4] \\ S_{10} &= 5[14 + 36] \\ S_{10} &= 5 \times 50 = 250 \end{aligned}$$

Arithmetic Progressions — Sum of first n terms

18. **18.** If the 4^{th} term of an Arithmetic Progression is 14 and the 12^{th} term is 38, find the first term a.

Answer:

$$5$$

Solution:

1. Write equations based on $a_n = a + (n-1)d$: $a + 3d = 14$ and $a + 11d = 38$.
2. Subtract the two equations to find d and then solve for a.

Derivation:

$$\begin{aligned} (a + 11d) - (a + 3d) &= 38 - 14 \\ 8d &= 24 \\ d &= 3 \\ a + 3(3) &= 14 \\ a + 9 &= 14 \\ a &= 5 \end{aligned}$$

Arithmetic Progressions — nth term of an AP

19. **19.** Find the sum of all two-digit numbers which, when divided by 4, yield 1 as a remainder.

Answer:

$$1210$$

Solution:

1. Identify the sequence: The smallest two-digit number is 10. $13 \div 4 = 3$ remainder 1. So the first term $a = 13$.
2. Find the last term: The largest two-digit number is 99. $97 \div 4 = 24$ remainder 1. So the last term $l = 97$.
3. Find the number of terms n: Use $a_n = a + (n-1)d$. $97 = 13 + (n-1)4$.
4. Calculate the sum using $S_n = \frac{n}{2}(a + l)$.

Derivation:

$$97 = 13 + 4n - 4$$
$$\implies 88 = 4n$$
$$\implies n = 22. S_{22} = \frac{22}{2}(13 + 97) = 11 \times 110 = 1210.$$

Arithmetic Progressions — Sum of n terms

20. 20. If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find the 16^{th} term of the AP.

Answer:

$$98$$

Solution:

1. Use the relationship between the n^{th} term and the sum of terms: $a_n = S_n - S_{n-1}$.
2. Calculate $S_{16} = 3(16)^2 + 5(16)$.
3. Calculate $S_{15} = 3(15)^2 + 5(15)$.
4. Subtract S_{15} from S_{16} to find a_{16}.

Derivation:

$$S_{16} = 3(256) + 80 = 768 + 80 = 848. S_{15} = 3(225) + 75 = 675 + 75 = 750. a_{16} = 848 - 750 = 98.$$

Arithmetic Progressions — Sum of n terms

21. 21. How many terms of the AP $24, 21, 18, \ldots$ must be taken so that their sum is 78?

Answer:

$$4 or 13$$

Solution:

1. Identify $a = 24$, $d = -3$, and $S_n = 78$.
2. Use the formula $S_n = \frac{n}{2}[2a + (n-1)d]$.
3. Solve the resulting quadratic equation for n.

4. Verify both values of n.

Derivation:

$$\begin{aligned}78 &= \frac{n}{2}[48 + (n-1)(-3)] \\ \implies 156 &= n(48 - 3n + 3) \\ \implies 156 &= 51n - 3n^2 \\ \implies 3n^2 - 51n + 156 &= 0 \\ \implies n^2 - 17n + 52 &= 0 \\ \implies (n-4)(n-13) &= 0.\end{aligned}$$

Arithmetic Progressions — Sum of n terms

22. 22. Find the middle term of the AP: $6, 13, 20, \ldots, 216$.

Answer:

$$111$$

Solution:

1. Find the total number of terms n using $a_n = a + (n-1)d$.
2. Identify the middle term position: If n is odd, middle term is $\frac{n+1}{2}$.
3. Calculate the value of the middle term.

Derivation: 216 = 6 + (n-1)7 $\implies$ $210 = 7(n-1)$ $\implies$ $30 = n - 1$ $\implies$ $n = 31$. Middle term is $\frac{31+1}{2} = 16^{th}$ term. a_{16} = 6 + 15(7) = 6 + 105 = 111 . *Arithmetic Progressions — n-th term*

23. 23. The 4^{th} term of an AP is 11 and the sum of its 5^{th} and 7^{th} terms is 34. Find the common difference.

Answer:

$$3$$

Solution:

1. Express a_4 as $a + 3d = 11$.
2. Express $a_5 + a_7$ as $(a + 4d) + (a + 6d) = 34$.
3. Solve the system of two linear equations for d.

Derivation: a + 3d = 11 (Eq 1). 2a + 10d = 34 $\implies$ $a+5d = 17$(Eq 2). Subtracting Eq 1 from Eq 2: (a + 5d) - (a + 3d) = 17 - 11 $\implies$ $2d = 6$ $\implies$ $d = 3$. *Arithmetic Progressions — n-th term*

24. 24. A sum of Rs. 2800 is to be used to award four prizes to students of a school for their overall academic performance. If each prize is Rs. 200 less than its preceding prize, find the value of each of the prizes.

Answer:

$$\text{Rs. } 850, \text{Rs. } 650, \text{Rs. } 450, \text{Rs. } 350$$

Solution:

1. Let the four prizes be $a+3d, a+d, a-d, a-3d$ or $a, a-d, a-2d, a-3d$. Let's use $a, a-d, a-2d, a-3d$.
2. The sum is $a+(a-d)+(a-2d)+(a-3d)=2800$.
3. Simplify the sum: $4a-6d=2800$. Given $d=200$.
4. Substitute $d=200$ into the equation: $4a-6(200)=2800$.
5. Solve for a: $4a-1200=2800 \Rightarrow 4a=4000 \Rightarrow a=1000$.
6. Calculate the prizes: $1000, 800, 600, 400$ is incorrect based on the sum logic. Correcting: Let prizes be $x, x-200, x-400, x-600$. Sum: $4x-1200=2800 \Rightarrow 4x=4000 \Rightarrow x=1000$. Prizes: $1000, 800, 600, 400$ is a sum of 2800. Wait, $1000+800+600+400=2800$. Correct.

Derivation:

$$\begin{aligned} S_n &= \frac{n}{2}[2a+(n-1)d] \\ \Rightarrow 2800 &= \frac{4}{2}[2a+3(-200)] \\ \Rightarrow 1400 &= 2a-600 \\ \Rightarrow 2a &= 2000 \\ \Rightarrow a &= 1000. \text{Prizes are} 1000, 800, 600, 400. \end{aligned}$$

Arithmetic Progressions — Sum of n terms of an AP

25. 25. The sum of the first n terms of an AP is given by $S_n = 3n^2+5n$. Find the 16^{th} term of this AP and the general expression for the n^{th} term.

Answer:

$$a_{16} = 92, a_n = 6n+2$$

Solution:

1. Use the formula $a_n = S_n - S_{n-1}$ for $n>1$.
2. Calculate $S_n = 3n^2+5n$.
3. Calculate $S_{n-1} = 3(n-1)^2+5(n-1) = 3(n^2-2n+1)+5n-5 = 3n^2-6n+3+5n-5 = 3n^2-n-2$.
4. Find $a_n = (3n^2+5n)-(3n^2-n-2) = 6n+2$.
5. Calculate $a_{16} = 6(16)+2 = 96+2 = 98$.

Derivation:

$$a_n = S_n - S_{n-1} = 3n^2+5n-[3(n-1)^2+5(n-1)] = 3n^2+5n-[3n^2-6n+3+5n-5] = 6n+2. \text{For n} = 16, a_{16} = 6(16)+2 = 98.$$

Arithmetic Progressions — Sum of n terms

26. **26.** In an AP, the sum of first p terms is q and the sum of first q terms is p. Show that the sum of first $(p+q)$ terms is $-(p+q)$.

Answer:

$$S_{p+q} = -(p+q)$$

Solution:

1. Write sum formulas: $S_p = \frac{p}{2}[2a+(p-1)d] = q$ and $S_q = \frac{q}{2}[2a+(q-1)d] = p$.
2. Simplify to $2a+(p-1)d = \frac{2q}{p}$ and $2a+(q-1)d = \frac{2p}{q}$.
3. Subtract equations to find d: $(p-q)d = \frac{2q}{p} - \frac{2p}{q} = \frac{2(q^2-p^2)}{pq} = \frac{-2(p-q)(p+q)}{pq}$.
4. Thus $d = \frac{-2(p+q)}{pq}$.
5. Find $2a$ and then $S_{p+q} = \frac{p+q}{2}[2a+(p+q-1)d]$.

Derivation: Subtracting gives d = -2/pq(p+q) . Substituting d back gives 2a = 2/p(q) - (p-1)d . Sum S_{p+q} = $\frac{p+q}{2}$[2a + (p+q-1)d] = -(p+q) . *Arithmetic Progressions — Sum of n terms*

27. **27.** A spiral is made up of successive semicircles, with centers alternately at A and B, starting with center at A of radii $0.5cm, 1.0cm, 1.5cm, 2.0cm, ...$ What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = \frac{22}{7}$)

Answer:

$$143cm$$

Solution:

1. The length of a semicircle is $L = \pi r$.
2. The lengths are $L_1 = 0.5\pi, L_2 = 1.0\pi, L_3 = 1.5\pi, \ldots$.
3. This forms an AP with $a = 0.5$ and $d = 0.5$.
4. We need the sum of $n = 13$ terms: $S_{13} = \frac{13}{2}[2(0.5\pi) + (13-1)(0.5\pi)]$.
5. Calculate $S_{13} = \frac{13}{2}[\pi + 6\pi] = \frac{13}{2}[7\pi] = \frac{91\pi}{2}$.
6. Substitute $\pi = \frac{22}{7}$: $S_{13} = \frac{91}{2} \times \frac{22}{7} = 13 \times 11 = 143$.

Derivation:

$$S_{13} = \frac{13}{2}[2(0.5\pi) + 12(0.5\pi)] = \frac{13}{2}[\pi + 6\pi] = \frac{13 \times 7 \times \pi}{2} = \frac{91}{2} \times \frac{22}{7} = 143.$$

Arithmetic Progressions — Applications of AP

28. **28.** In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2$ cm, $DB = 3$ cm, and $AE = 3$ cm. Find the length of EC.

Answer: (C)

Solution:

1. Apply Thales Theorem (Basic Proportionality Theorem) as $DE \parallel BC$.

2. The ratio $\frac{AD}{DB} = \frac{AE}{EC}$ must hold.
3. Substitute the given values: $\frac{2}{3} = \frac{3}{EC}$.
4. Solve for EC: $EC = \frac{3\times 3}{2} = 4.5$ cm.

Derivation:

$$\begin{aligned} \frac{AD}{DB} &= \frac{AE}{EC} \\ \implies \frac{2}{3} &= \frac{3}{EC} \\ \implies 2EC &= 9 \\ \implies EC &= 4.5 \end{aligned}$$

Triangles — Basic Proportionality Theorem

29. 29. If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio of their areas $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)}$ is:

Answer: (B)

Solution:

1. Recall the theorem: The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
2. Calculate the square of the given ratio: $(\frac{1}{3})^2$.
3. Result is $\frac{1}{9}$.

Derivation:

$$\frac{\text{Area}(ABC)}{\text{Area}(PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Triangles — Areas of Similar Triangles

30. 30. In $\triangle ABC$, if $AB = 6\sqrt{3}$ cm, $AC = 12$ cm, and $BC = 6$ cm, then the measure of $\angle B$ is:

Answer: (D)

Solution:

1. Check if the sides satisfy the Converse of Pythagoras Theorem: $AB^2 + BC^2 = AC^2$.
2. Calculate squares: $(6\sqrt{3})^2 = 36 \times 3 = 108$, $6^2 = 36$, $12^2 = 144$.
3. Observe $108 + 36 = 144$. Since $AB^2 + BC^2 = AC^2$, the triangle is right-angled at B.

Derivation:

$$(6\sqrt{3})^2 + 6^2 = 108 + 36 = 144 = 12^2$$

Triangles — Pythagoras Theorem

31. **31.** Two triangles are similar if their corresponding sides are in the same ratio. This criterion is known as:

Answer: (A)

Solution:

1. SSS (Side-Side-Side) similarity criterion states that if the sides of one triangle are proportional to the sides of another triangle, then the triangles are similar.

Derivation:

$$\text{If } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}, \text{ then } \triangle ABC \sim \triangle PQR \text{ (SSS Similarity)}$$

Triangles — Criteria for Similarity

32. **32.** The altitude of an equilateral triangle with side $2a$ is:

Answer: (C)

Solution:

1. In an equilateral triangle, the altitude bisects the base.
2. Using Pythagoras theorem in the half-triangle: $h^2 + a^2 = (2a)^2$.
3. Solve for h: $h^2 = 4a^2 - a^2 = 3a^2 \implies h = a\sqrt{3}$.

Derivation:

$$h = \sqrt{(2a)^2 - a^2} = \sqrt{4a^2 - a^2} = \sqrt{3a^2} = a\sqrt{3}$$

Triangles — Properties of Triangles

33. **33.** Assertion (A): If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Reason (R): This is the Converse of Basic Proportionality Theorem.

Answer: (A)

Solution:

1. Identify the statement as the definition of the converse of Thales Theorem (BPT).
2. Confirm that the reason correctly identifies the theorem name.

Derivation:

$$A : \text{Converse of BPT holds true. } R : \text{Definition of Converse of BPT.}$$

Triangles — Basic Proportionality Theorem

34. **34.** Assertion (A): Two triangles are similar if their corresponding angles are equal. Reason (R): If two triangles are equiangular, they are similar by AAA similarity criterion.

Answer: (A)

Solution:

1. Verify the AAA similarity criterion.
2. Check if equality of angles leads to similarity.

Derivation:

$$A : \text{Angle-Angle-Angle Similarity Criterion } R : \text{AAA creates proportional sides.}$$

Triangles — Similarity of Triangles

35. **35.** Assertion (A): In $\triangle ABC$, if $DE \parallel BC$ such that $AD/DB = 1/2$ and $AE = 2$ cm, then $EC = 4$ cm. Reason (R): By Basic Proportionality Theorem, if $DE \parallel BC$, then $AD/DB = AE/EC$.

Answer: (A)

Solution:

1. Apply BPT: $1/2 = 2/EC$.
2. Solve for EC: $EC = 4$ cm.

Derivation:

$$\begin{aligned} \frac{AD}{DB} &= \frac{AE}{EC} \\ \implies \frac{1}{2} &= \frac{2}{EC} \\ \implies EC &= 4 \end{aligned}$$

Triangles — Basic Proportionality Theorem

36. **36.** Assertion (A): All congruent triangles are similar. Reason (R): All similar triangles are congruent.

Answer: (C)

Solution:

1. Congruent triangles have same shape and size, so they are similar.
2. Similar triangles have same shape but not necessarily same size, so they are not always congruent.

Derivation:

$$\text{Congruence} \implies \text{Similarity (True). SimilarityRightarrowCongruence (False).}$$

Triangles — Similarity vs Congruence

37. **37.** Assertion (A): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Reason (R): This is the Area of Similar Triangles Theorem.

Answer: (A)

Solution:

1. Recall the theorem regarding areas of similar triangles.
2. Confirm that the statement matches the standard theorem.

Derivation:

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$

Triangles — Areas of Similar Triangles

38. 38. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm. Find the length of DB.

Answer:

3.6 cm

Solution:

1. Apply Basic Proportionality Theorem (Thales Theorem).
2. Substitute the given values and solve for DB.

Derivation:

$$\begin{aligned} \frac{AD}{DB} &= \frac{AE}{EC} \\ \implies \frac{2.4}{DB} &= \frac{3.2}{4.8} \\ \implies DB = \frac{2.4 \times 4.8}{3.2} &= 3.6 \text{ cm} \end{aligned}$$

Triangles — Basic Proportionality Theorem

39. 39. Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If Area($\triangle ABC$) $= 64$ cm^2 and Area($\triangle DEF$) $= 121$ cm^2, and $EF = 15.4$ cm, find the length of side BC.

Answer:

11.2 cm

Solution:

1. Use the theorem that the ratio of areas of similar triangles equals the square of the ratio of their corresponding sides.
2. Substitute values and take the square root to find BC.

Derivation:

$$\begin{aligned} \frac{\text{Area}(ABC)}{\text{Area}(DEF)} &= \left(\frac{BC}{EF}\right)^2 \\ \implies \frac{64}{121} &= \left(\frac{BC}{15.4}\right)^2 \\ \implies \frac{8}{11} &= \frac{BC}{15.4} \\ \implies BC = \frac{8 \times 15.4}{11} &= 11.2 \text{ cm} \end{aligned}$$

40. **40.** A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

Answer:

$$42 \text{ m}$$

Solution:

1. Recognize that the triangles formed by the object and shadow are similar by AA similarity.
2. Set up the ratio of corresponding sides and solve for the height.

Derivation:

$$\frac{6}{h} = \frac{4}{28}$$
$$\implies h = \frac{6 \times 28}{4} = 6 \times 7 = 42 \text{ m}$$

Triangles — Similar Triangles